Retirement Plan Access and Participation Among Private-Sector Workers, 2010–2024

Alejandro De La Torre

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1 Introduction

This exercise constructs annual measures of retirement plan *access* and *participation* among private-sector wage-and-salary workers ages 20–64 using the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) from 2010 through 2024. Access refers to whether an employer offers a retirement plan, while participation refers to whether the worker is included in that plan. Using IPUMS-CPS microdata, I construct weighted annual estimates and document the sample construction and measurement choices used to produce a transparent and replicable time series of retirement plan coverage.

2 Sample Construction

The analysis uses microdata from the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) for survey years 2010 through 2024. The unit of observation is the individual respondent. To construct the analytic sample, I apply a series of restrictions that align the sample with the population of interest: private-sector workers of working age.

First, the sample is restricted to individuals ages 20–64 using the variable `AGE`. This restriction focuses the analysis on the working-age population while excluding younger respondents who are typically not fully attached to the labor force and older individuals who are more likely to be retired.

Second, I restrict the sample to individuals who are currently employed using the CPS employment-status variable `EMPSTAT`. Respondents coded as `EMPSTAT = 10` ("at work") or `EMPSTAT = 12` ("has job, not at work last week") are classified as employed and retained in the sample. Individuals who are unemployed or not in the labor force are excluded.

Third, among employed individuals I restrict the sample to private-sector wage-and-salary workers using the class-of-worker variable `CLASSWKR`. In this extract, the relevant category corresponds to code 21, which represents private wage-and-salary workers. Self-employed individuals, government workers, unpaid family workers, and other employment categories are excluded. This restriction ensures that the denominator corresponds to workers who could plausibly receive an employer-sponsored retirement plan from a private-sector employer.

After applying these restrictions, the resulting analytic sample consists of private-sector wage-and-salary workers ages 20–64 observed in the CPS ASEC between 2010 and 2024.

3 Measurement of Retirement Plan Access and Participation

Retirement plan access and participation are measured using CPS ASEC questions regarding employer-sponsored pension or retirement plans. The analysis distinguishes between two related but conceptually distinct outcomes: whether a worker has access to a retirement plan offered by their employer and whether the worker actually participates in such a plan.

Access is measured using the variable `UH_PENPLAN_A1`, which indicates whether the respondent's employer offers a pension or retirement plan for any of its employees. In the CPS questionnaire, respondents are asked whether their employer provides such a plan. Workers who respond affirmatively are coded as having access to a retirement plan. In the analysis, access is coded as a binary indicator equal to 1 if `UH_PENPLAN_A1 = 1` and 0 if `UH_PENPLAN_A1 = 2`.

Participation is measured using two CPS variables. First, the variable `UH_PENSINCL_A1` indicates whether the respondent is included in the employer's pension or retirement plan. This variable is used to construct a conditional participation measure among workers whose employers offer a plan. Conditional participation is coded as 1 if `UH_PENSINCL_A1 = 1` and 0 if `UH_PENSINCL_A1 = 2`.

Second, the variable `PENSION` provides an additional indicator of whether the worker is covered by a pension plan at work. This variable is used as a cross-check when constructing an unconditional participation measure. In the unconditional measure, participation is coded as 1 if `PENSION = 3` and 0 otherwise.

Together, these variables allow the analysis to distinguish between workers whose employers offer retirement plans and workers who are actually enrolled in those plans.

4 Weighting

All estimates are calculated using the CPS ASEC person weight `ASECWT`. The CPS ASEC is a complex survey designed to represent the U.S. population, and the person weight adjusts for the survey's sampling design, differential probabilities of selection, and nonresponse. Applying `ASECWT` ensures that computed statistics represent population-level estimates rather than simple averages from the raw survey sample.

In practice, the weight is applied when computing all annual means and proportions. As a result, reported access and participation rates should be interpreted as estimates for the national population of private-sector wage-and-salary workers ages 20–64.

5 Treatment of Missing or Ambiguous Responses

The CPS variables used in the analysis contain categories for responses that are not in universe or otherwise missing. In constructing the outcome measures, responses coded as "not in universe" are treated as zero when calculating weighted means for the full analytic sample. This approach preserves a consistent denominator when computing access and participation rates.

For example, when constructing the access variable from `UH_PENPLAN_A1`, responses indicating that a worker's employer does not offer a retirement plan are coded as 0, while affirmative responses are coded as 1. Observations coded as not in universe are treated as zero in the weighted calculation so that workers without access remain part of the denominator.

Similarly, when constructing participation measures, responses indicating nonparticipation are coded as 0, while affirmative responses are coded as 1. Conditional participation is evaluated among workers with access to an employer-sponsored plan, while the unconditional measure uses the broader pension coverage variable for validation and consistency checks.

These choices follow the logic implemented in the analysis code and ensure that annual estimates are constructed consistently across survey years.

6 Implications of the 2015 CPS ASEC Redesign

In 2015 the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) implemented a redesigned questionnaire and processing system intended to improve the measurement of income and program participation. The redesign introduced new income questions, revised survey sequencing, and updated data processing methods. As a result, several income-related measures in the CPS ASEC exhibit structural differences beginning with the 2015 survey.

Although retirement plan access and participation are not direct income measures, the redesign can still affect interpretation of time-series estimates. Changes in questionnaire structure and survey administration may alter how respondents understand or answer questions about employer benefits, including pension and retirement plan coverage. Consequently, observed differences in retirement plan access or participation around 2015 may partly reflect measurement changes rather than purely economic or institutional trends.

For this reason, estimates spanning the 2010–2024 period should be interpreted with awareness of the 2015 CPS ASEC redesign. When analyzing trends across this boundary, it is important to consider the possibility of a break in series driven by survey methodology rather than underlying changes in retirement plan coverage.

A Appendix Figures

A.1 Main Figure

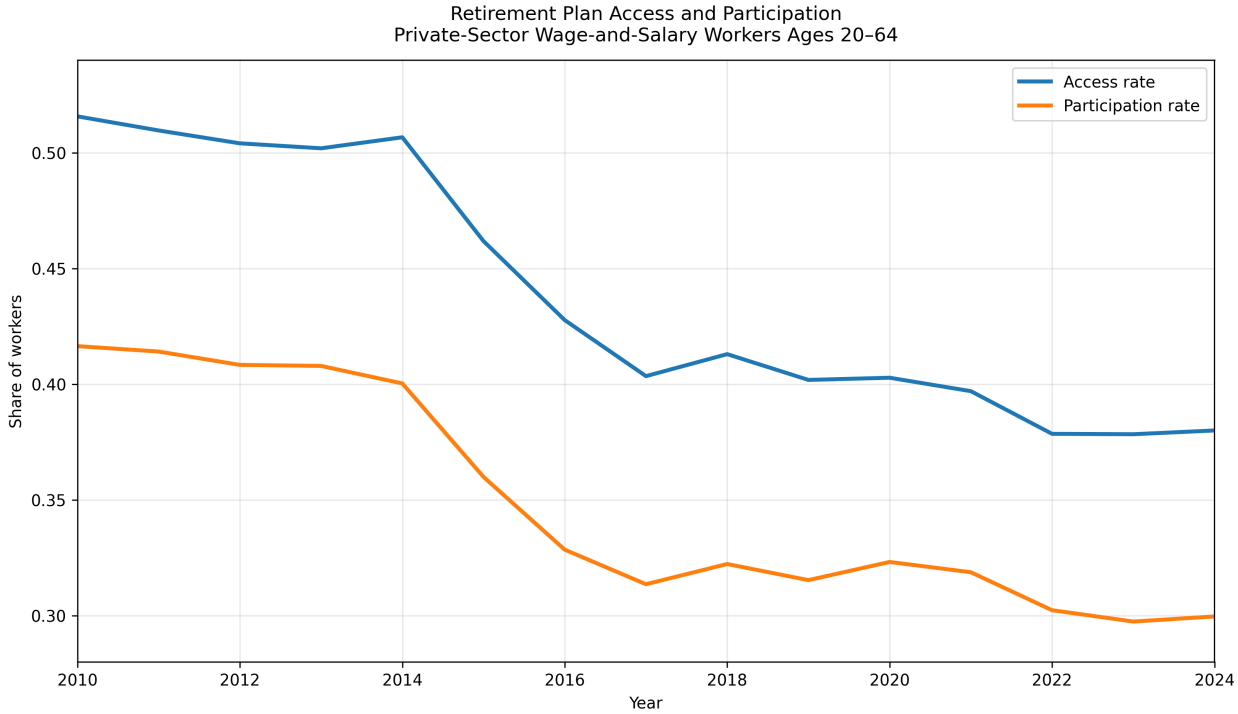

Figure 1: Retirement plan access and participation rates among private-sector wage-and-salary workers ages 20–64, 2010–2024.

This figure plots two weighted annual series for private-sector wage-and-salary workers ages 20–64 who are currently employed. For each year, the access rate is calculated as the weighted share of workers whose employer offers a retirement plan, using `UH_PENPLAN_A1`. The participation rate shown here is the unconditional participation rate: the weighted share of workers in the analytic sample who are included in a retirement plan, using `PENSION`. In both cases, the annual rate is computed with the CPS ASEC person weight (`ASECWT`), so each point represents a population-level estimate for that survey year rather than a raw sample mean.

Mechanically, the figure plots one weighted access rate and one weighted participation rate for each year from 2010 through 2024. Access is higher than participation because the two measures capture different margins: access asks whether a worker's employer offers a plan at all, while participation asks whether the worker is actually covered by a plan. A worker can therefore have no opportunity to participate because no plan is offered, or may have access but still not be enrolled. The decline after 2014 should also be interpreted cautiously. Some of that movement may reflect real changes in employer-sponsored retirement coverage, but some may reflect the 2015 CPS ASEC redesign, which changed questionnaire structure and processing methods and could have affected measured levels around that break.

Source: IPUMS CPS, Current Population Survey Annual Social and Economic Supplement (ASEC), 2010–2024, `https://cps.ipums.org/cps/`. *Note:* Standard errors and confidence intervals were computed in the analysis workflow but are not displayed because they are visually negligible at this scale.

A.2 Yearly Annual Access Rate and Sample Size

Table 1 shows how the annual retirement plan access rate is constructed for the sample used in the analysis. In each year, the sample includes private-sector wage-and-salary workers ages 20–64 who are currently employed. The table reports the number of workers in that sample, the number whose employer offers a retirement plan, the number whose employer does not offer a plan, and the small number of observations for which access information is missing.

This construction is useful because the quantity of interest is the share of workers represented in the population—not just the share of respondents in the sample—whose employer offers a retirement plan in a given year. An alternative would have been to report an unweighted access rate using only the raw sample counts, or to report a simple average of the annual access rates across years. I do not use those approaches here. Instead, each year's access rate is calculated as the ratio of two weighted population counts for that year: the estimated number of workers whose employer offers a plan divided by the estimated number of workers with nonmissing access information. The pooled row at the bottom of the table is not the simple average of the annual rates. Rather, it is constructed by summing the weighted numerators across all years, summing the weighted denominators across all years, and then taking the ratio of those two totals. It therefore represents the overall weighted access rate for the pooled 2010–2024 sample.

Table 1: Annual retirement plan access rate construction, CPS ASEC 2010–2024

Year	Sample workers	Offers plan	No plan	Missing	Weighted offers	Weighted sample	Access (%)
2010	62,712	33,374	27,989	1,349	48.17	91.17	52.84
2011	61,968	32,410	28,108	1,450	48.50	92.72	52.31
2012	61,862	32,263	28,087	1,512	49.25	95.04	51.82
2013	62,212	32,132	28,630	1,450	49.50	96.12	51.50
2014	61,883	32,602	27,921	1,360	102.34	197.14	51.91
2015	62,005	29,000	31,562	1,443	47.33	99.89	47.39
2016	57,764	24,934	31,618	1,212	44.60	102.05	43.70
2017	58,682	24,190	33,246	1,246	42.64	103.36	41.25
2018	56,794	23,655	31,924	1,215	44.30	104.91	42.23
2019	57,631	23,494	32,917	1,220	43.63	106.12	41.11
2020	48,615	20,153	27,640	822	42.97	104.72	41.04
2021	49,141	19,976	28,201	964	41.06	101.26	40.55
2022	47,663	18,627	27,900	1,136	40.93	105.40	38.84
2023	45,960	17,909	26,994	1,057	41.69	107.38	38.82
2024	45,950	17,891	27,082	977	42.03	108.11	38.88
Total / pooled	840,842	382,610	439,819	18,413	728.95	1,615.38	45.13

Notes: "Sample workers" is the number of private-sector wage-and-salary workers ages 20–64 in the sample for that year. "Offers plan" is the unweighted count of workers whose employer offers a retirement plan; "No plan" is the unweighted count whose employer does not offer a retirement plan; and "Missing" is the unweighted count with missing access information. "Weighted offers" is the sum of CPS ASEC person weights for workers whose employer offers a plan, and "Weighted sample" is the sum of person weights over all workers with nonmissing access information. These weighted counts are interpreted as estimated population counts and are reported in millions. The access rate is calculated as

$$\widehat{Access}_t = \frac{\sum_i w_i \, \mathbf{1}\{access_i = 1\}}{\sum_i w_i \, \mathbf{1}\{access_i \in \{0,1\}\}} \times 100.$$

This calculation assumes that the CPS ASEC person weight can be used to scale each observed respondent up to the number of similar people they represent in the broader population, so that summing weights yields estimated population totals. The pooled row aggregates the weighted numerator and denominator across all years and reports the corresponding pooled access rate. *Source:* IPUMS CPS, Current Population Survey Annual Social and Economic Supplement (ASEC), 2010–2024, `https://cps.ipums.org/cps/`.

A.3 Variable Definitions and Coding

Table 2 documents the variables used to define the analytic sample, construct the main outcomes, and validate consistency checks in the Component 1 workflow.

Table 2: Variable definitions and coding for Component 1 analysis

Variable name	Role in analysis	Official label	Coding used in this project	Notes / interpretation
`YEAR`	Sample restriction; grouping	Survey year	Keep 2010–2024; group by year	Used to construct annual estimates
`AGE`	Sample restriction	Age	Keep 20–64	Working-age adults
`EMPSTAT`	Sample restriction	Employment status	Keep 10 (at work) and 12 (has job, not at work last week)	Excludes unemployed and NILF
`CLASSWKR`	Sample restriction	Class of worker	Keep 21 (private wage/salary)	Other private codes are not observed in this extract
`ASECWT`	Weighting	ASEC person weight	Used as person-level weight in all annual estimates	Weights used in means and SEs
`UH_PENPLAN_A1`	Access construction	Existence of pension plan by employer, 1980–2025	`access`=1 if 1; `access`=0 if 2	NIU treated as zero in weighted means
`UH_PENSINCL_A1`	Conditional participation	Included in pension plan at work, 1980–2025	`participation_conditional`=1 if 1; =0 if 2	Computed among `access`=1
`PENSION`	Unconditional participation; validation	Pension plan at work	`participation_unconditional`=1 if 3; =0 if 1 or 2	Cross-checks access/participation coding
`access`	Derived outcome	Constructed	Derived from `UH_PENPLAN_A1`	Used for access rate and denominator for conditional participation
`participation_unconditional`	Derived outcome	Constructed	Derived from `PENSION`	Used in final tables and figures
`participation_conditional`	Derived outcome	Constructed	Derived from `UH_PENSINCL_A1`	Used in final tables and figures